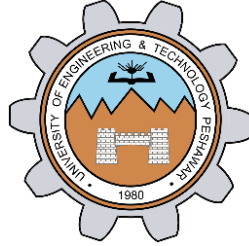


Use Wireshark to View Network Traffic Topology

LAB # 04



Spring 2024

CSE-303L Data Communication & Networks Lab

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Registration No.: **21PWCSE2059**

Class Section: **C**

“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

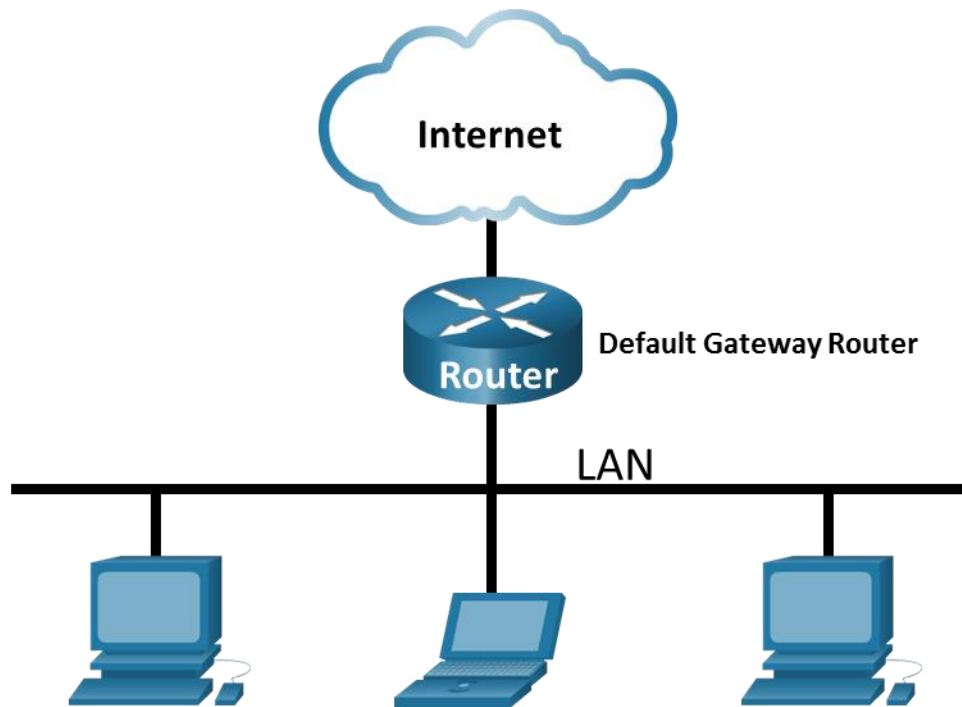
Submitted to:

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Date:

27th March 2024

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Objectives

Part 1: Capture and Analyze Local ICMP Data in Wireshark

Part 2: Capture and Analyze Remote ICMP Data in Wireshark

Background / Scenario

Wireshark is a software protocol analyzer, or "packet sniffer" application, used for network troubleshooting, analysis, software and protocol development, and education. As data streams travel back and forth over the network, the sniffer "captures" each protocol data unit (PDU) and can decode and analyze its content according to the appropriate RFC or other specifications.

Wireshark is a useful tool for anyone working with networks and can be used with most labs for data analysis and troubleshooting. In this lab, you will use Wireshark to capture ICMP data packet IP addresses and MAC addresses.

Required Resources

- 1 PC (Windows with internet access)
- Additional PCs on a local-area network (LAN) will be used to reply to ping requests.

Part 1: Capture and Analyze Local ICMP Data in Wireshark

Questions:

```
C:\Users\aalaa>ping 192.168.105.193

Pinging 192.168.105.193 with 32 bytes of data:
Reply from 192.168.105.193: bytes=32 time=90ms TTL=128
Reply from 192.168.105.193: bytes=32 time=89ms TTL=128
Reply from 192.168.105.193: bytes=32 time=78ms TTL=128
Reply from 192.168.105.193: bytes=32 time=105ms TTL=128

Ping statistics for 192.168.105.193:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 78ms, Maximum = 105ms, Average = 90ms
```

No.	Time	Source	Destination	Protocol	Length	Info
13	0.665710	178.60.13.20	192.168.105.216	ICMP		70 Destination unreachable (Port unreachable)
14	0.714916	46.150.78.0	192.168.105.216	ICMP		70 Destination unreachable (Port unreachable)
126	9.337853	213.59.128.134	192.168.105.216	ICMP		94 Destination unreachable (Network unreachable)
139	10.384603	213.59.128.134	192.168.105.216	ICMP		94 Destination unreachable (Network unreachable)
157	12.051865	213.59.128.134	192.168.105.216	ICMP		94 Destination unreachable (Network unreachable)
210	15.942343	213.59.128.134	192.168.105.216	ICMP		94 Destination unreachable (Network unreachable)
320	22.980984	192.168.105.216	192.168.105.193	ICMP		74 Echo (ping) request id=0x0001, seq=45/11520, ttl=128 (reply in 321)
321	23.011273	192.168.105.193	192.168.105.216	ICMP		74 Echo (ping) reply id=0x0001, seq=45/11520, ttl=128 (request in 320)
336	23.725092	213.59.128.134	192.168.105.216	ICMP		94 Destination unreachable (Network unreachable)
337	24.004809	192.168.105.216	192.168.105.193	ICMP		74 Echo (ping) request id=0x0001, seq=46/11776, ttl=128 (reply in 338)
338	24.139823	192.168.105.193	192.168.105.216	ICMP		74 Echo (ping) reply id=0x0001, seq=46/11776, ttl=128 (request in 337)
350	25.026228	192.168.105.216	192.168.105.193	ICMP		74 Echo (ping) request id=0x0001, seq=47/12032, ttl=128 (reply in 351)
351	25.059368	192.168.105.193	192.168.105.216	ICMP		74 Echo (ping) reply id=0x0001, seq=47/12032, ttl=128 (request in 350)

> Frame 350: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF{...} ...

▼ Ethernet II, Src: LiteonTechno_ba:0b:bd (cc:b0:da:ba:0b:bd), Dst: AzureWaveTec_2e:1a:15 (20:4e:f6:2e:1a:15)

 Destination: AzureWaveTec_2e:1a:15 (20:4e:f6:2e:1a:15)

 Address: AzureWaveTec_2e:1a:15 (20:4e:f6:2e:1a:15)

 = LG bit: Globally unique address (factory default)

 = IG bit: Individual address (unicast)

 Source: LiteonTechno_ba:0b:bd (cc:b0:da:ba:0b:bd)

 Address: LiteonTechno_ba:0b:bd (cc:b0:da:ba:0b:bd)

 = LG bit: Globally unique address (factory default)

 = IG bit: Individual address (unicast)

 Type: IPv4 (0x0800)

> Internet Protocol Version 4, Src: 192.168.105.216, Dst: 192.168.105.193

> Internet Control Message Protocol

Does the source MAC address match your PC interface?

Ans: Yes

```
▼ Ethernet II, Src: LiteonTechno_ba:0b:bd (cc:b0:da:ba:0b:bd), Dst: AzureWaveTec_2e:1a:15 (20:4e:f6:2e:1a:15)
    > Destination: AzureWaveTec_2e:1a:15 (20:4e:f6:2e:1a:15)
    > Source: LiteonTechno_ba:0b:bd (cc:b0:da:ba:0b:bd)
    Type: IPv4 (0x0800)
```

```
Connection-specific DNS Suffix . : 
Description . . . . . : Dell Wireless 1820A 802.11ac
Physical Address. . . . . : CC-B0-DA-BA-0B-BD
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

Does the destination MAC address in Wireshark match your team member MAC address?

Ans: Yes. It is **20:4e:f6:2e:1a:15** (team member's MAC address).

How is the MAC address of the pinged PC obtained by your PC?

Ans: The ping command does not directly obtain the MAC address of the target device. However, once the ICMP echo request reaches the local network, the Address Resolution Protocol (ARP) is used to obtain the MAC address associated with the IP address. The ARP protocol broadcasts a request for the MAC address associated with the IP address being pinged, and the device with that IP address responds with its MAC address. This allows the ping command to send packets to the correct MAC address for the target device.

Part 2: Capture and Analyze Remote ICMP Data in Wireshark

List the destination IP and MAC addresses for all three locations in the space provided.

IP address for **www.yahoo.com**:

Ans: **69.147.88.8**

```
C:\Users\aaliala>ping -4 www.yahoo.com

Pinging me-ycpi-cf-www.g06.yahoodns.net [69.147.88.8] with 32 bytes of data:
Reply from 69.147.88.8: bytes=32 time=435ms TTL=48
Reply from 69.147.88.8: bytes=32 time=375ms TTL=48
Reply from 69.147.88.8: bytes=32 time=343ms TTL=48
Reply from 69.147.88.8: bytes=32 time=444ms TTL=48

Ping statistics for 69.147.88.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 343ms, Maximum = 444ms, Average = 399ms

C:\Users\aaliala>
```

MAC address for **www.yahoo.com**:

Ans: **7a:b3:56:d0:a3:ad**(MAC Address of Router)

59444	853.032294	192.168.114.216	69.147.88.8	ICMP	74 Echo (ping) request	id=0x0001, seq=60/15360, ttl=128 (reply in 59480)
59480	853.467748	69.147.88.8	192.168.114.216	ICMP	74 Echo (ping) reply	id=0x0001, seq=60/15360, ttl=48 (request in 59444)
59534	854.046474	192.168.114.216	69.147.88.8	ICMP	74 Echo (ping) request	id=0x0001, seq=61/15616, ttl=128 (reply in 59560)
59560	854.422278	69.147.88.8	192.168.114.216	ICMP	74 Echo (ping) reply	id=0x0001, seq=61/15616, ttl=48 (request in 59534)
59632	855.061827	192.168.114.216	69.147.88.8	ICMP	74 Echo (ping) request	id=0x0001, seq=62/15872, ttl=128 (reply in 59672)
59672	855.404419	69.147.88.8	192.168.114.216	ICMP	74 Echo (ping) reply	id=0x0001, seq=62/15872, ttl=48 (request in 59632)
59722	856.086407	192.168.114.216	69.147.88.8	ICMP	74 Echo (ping) request	id=0x0001, seq=63/16128, ttl=128 (reply in 59744)
59744	856.530727	69.147.88.8	192.168.114.216	ICMP	74 Echo (ping) reply	id=0x0001, seq=63/16128, ttl=48 (request in 59722)

```
> Frame 59444: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface ^
^ Ethernet II, Src: LiteonTechno_ba:0b:bd (cc:b0:da:ba:0b:bd), Dst: 7a:b3:56:d0:a3:ad
  Destination: 7a:b3:56:d0:a3:ad (7a:b3:56:d0:a3:ad)
    Address: 7a:b3:56:d0:a3:ad (7a:b3:56:d0:a3:ad)
      .... 1. .... = LG bit: Locally administered address (this is
      .... 0 .... = IG bit: Individual address (unicast)
    > Source: LiteonTechno_ba:0b:bd (cc:b0:da:ba:0b:bd)
      Type: IPv4 (0x0800)
  > Internet Protocol Version 4, Src: 192.168.114.216, Dst: 69.147.88.8
  > Internet Control Message Protocol
```

IP address for www.cisco.com:

Ans: 23.52.114.126

```
C:\Users\aalialia>ping -4 www.cisco.com

Pinging e2867.dsca.akamaiedge.net [23.52.114.126] with 32 bytes of data:
Reply from 23.52.114.126: bytes=32 time=240ms TTL=51
Reply from 23.52.114.126: bytes=32 time=147ms TTL=51
Reply from 23.52.114.126: bytes=32 time=153ms TTL=51
Reply from 23.52.114.126: bytes=32 time=149ms TTL=51

Ping statistics for 23.52.114.126:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 147ms, Maximum = 240ms, Average = 172ms

C:\Users\aalialia>
```

MAC address for www.cisco.com:

Ans: 7a:b3:56:d0:a3:ad(MAC Address of Router)

```
86671 1257.875995 192.168.114.216 23.52.114.126 ICMP 74 Echo (ping) request id=0x0001, seq=64/16384, ttl=128 (reply in 86680)
86680 1258.116349 23.52.114.126 192.168.114.216 ICMP 74 Echo (ping) reply id=0x0001, seq=64/16384, ttl=51 (request in 86671)
86687 1258.221397 176.37.198.162 192.168.114.216 ICMP 94 Destination unreachable (Host unreachable)
86688 1258.221397 176.37.198.162 192.168.114.216 ICMP 94 Destination unreachable (Host unreachable)
86751 1258.887289 192.168.114.216 23.52.114.126 ICMP 74 Echo (ping) request id=0x0001, seq=65/16640, ttl=128 (reply in 86759)
86759 1259.034688 23.52.114.126 192.168.114.216 ICMP 74 Echo (ping) reply id=0x0001, seq=65/16640, ttl=51 (request in 86751)
86832 1259.906921 192.168.114.216 23.52.114.126 ICMP 74 Echo (ping) request id=0x0001, seq=66/16896, ttl=128 (reply in 86834)
86834 1260.059825 23.52.114.126 192.168.114.216 ICMP 74 Echo (ping) reply id=0x0001, seq=66/16896, ttl=51 (request in 86832)
86907 1260.926315 192.168.114.216 23.52.114.126 ICMP 74 Echo (ping) request id=0x0001, seq=67/17152, ttl=128 (reply in 86912)
86912 1261.075038 23.52.114.126 192.168.114.216 ICMP 74 Echo (ping) reply id=0x0001, seq=67/17152, ttl=51 (request in 86907)
87019 1262.799904 176.37.198.162 192.168.114.216 ICMP 94 Destination unreachable (Host unreachable)
87515 1269.967816 176.37.198.162 192.168.114.216 ICMP 94 Destination unreachable (Host unreachable)
87563 1270.934073 188.233.116.103 192.168.114.216 ICMP 90 Destination unreachable (Host unreachable)

> Frame 86671: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface ^
Ethernet II, Src: LiteonTechno_ba:0b:bd (cc:b0:da:ba:0b:bd), Dst: 7a:b3:56:d0:a3:ad
  Destination: 7a:b3:56:d0:a3:ad (7a:b3:56:d0:a3:ad)
    Address: 7a:b3:56:d0:a3:ad (7a:b3:56:d0:a3:ad)
      ...1. .... = LG bit: Locally administered address (this is
      ...0. .... = IG bit: Individual address (unicast)
  Source: LiteonTechno_ba:0b:bd (cc:b0:da:ba:0b:bd)
    Address: LiteonTechno_ba:0b:bd (cc:b0:da:ba:0b:bd)
      ...0. .... = LG bit: Globally unique address (factory default)
      ...1. .... = IG bit: Individual address (unicast)

0000 7a b3 56 d0 a3 ac cc b0 da ba 0b bd 08 00 45 00 z-V... ..E-
0010 00 3c 76 86 00 00 00 01 07 08 c0 a8 72 d8 17 34 <V.....p..4
0020 72 7e 08 00 4d 1b 00 01 00 40 61 62 63 64 65 66 ~..M...@abcdef
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklm opqrstuv
0040 77 61 62 63 64 65 66 67 68 69 ..wabcdefg hi
```

IP address for www.google.com:

Ans: 172.217.17.36

```
C:\Users\aalialia>ping -4 www.google.com

Pinging www.google.com [172.217.17.36] with 32 bytes of data:
Reply from 172.217.17.36: bytes=32 time=72ms TTL=54
Reply from 172.217.17.36: bytes=32 time=74ms TTL=54
Reply from 172.217.17.36: bytes=32 time=68ms TTL=54
Reply from 172.217.17.36: bytes=32 time=64ms TTL=54

Ping statistics for 172.217.17.36:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 64ms, Maximum = 74ms, Average = 69ms

C:\Users\aalialia>
```

MAC address for www.google.com:

Ans: 7a:b3:56:d0:a3:ad(MAC Address of Router)

icmp						
No.	Time	Source	Destination	Protocol	Length	Info
1264	36.999544	95.183.43.15	192.168.114.216	ICMP	94	Destina
1547	45.009705	85.114.11.38	192.168.114.216	ICMP	94	Destina
1680	51.026618	192.168.114.216	172.217.17.36	ICMP	74	Echo (f
1683	51.098747	172.217.17.36	192.168.114.216	ICMP	74	Echo (f
1735	52.023683	95.183.43.15	192.168.114.216	ICMP	94	Destina
1736	52.044478	192.168.114.216	172.217.17.36	ICMP	74	Echo (f
1744	52.118568	172.217.17.36	192.168.114.216	ICMP	74	Echo (f
1808	53.064145	192.168.114.216	172.217.17.36	ICMP	74	Echo (f
1815	53.132504	172.217.17.36	192.168.114.216	ICMP	74	Echo (f
1879	54.088377	192.168.114.216	172.217.17.36	ICMP	74	Echo (f
1880	54.153032	172.217.17.36	192.168.114.216	ICMP	74	Echo (f
1905	55.212770	192.168.11.1	192.168.114.216	ICMP	70	Time-to
1935	56.214417	192.168.11.1	192.168.114.216	ICMP	70	Time-to
1985	58.214799	192.168.11.1	192.168.114.216	ICMP	70	Time-to
2223	62.222645	192.168.11.1	192.168.114.216	ICMP	70	Time-to
2646	70.236243	192.168.11.1	192.168.114.216	ICMP	70	Time-to
8806	104.094489	109.202.166.39	192.168.114.216	ICMP	94	Destina
14074	133.106077	103.123.226.242	192.168.114.216	ICMP	94	Time-to
14159	136.107268	103.123.226.242	192.168.114.216	ICMP	94	Time-to

> Frame 1736: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \De

> Ethernet II, Src: LiteonTechno_ba:0b:bd (cc:b0:da:ba:0b:bd), Dst: 7a:b3:56:d0:a3:ad (7a:b3:56:d0:a3:ad)

> Destination: 7a:b3:56:d0:a3:ad (7a:b3:56:d0:a3:ad)

> Source: LiteonTechno_ba:0b:bd (cc:b0:da:ba:0b:bd)

Type: IPv4 (0x0800)

> Internet Protocol Version 4, Src: 192.168.114.216, Dst: 172.217.17.36

> Internet Control Message Protocol

0000 7a

0010 00

0020 11

0030 67

0040 77

What is significant about this information?

Ans: The MAC addresses for all three locations are the same.

How does this information differ from the local ping information you received in Part 1?

Ans: A ping to a local host returns the MAC address of the PC NIC. A ping to a remote host returns the MAC address of the default gateway LAN interface.

Why does Wireshark show the actual MAC address of the local hosts, but not the actual MAC address for the remote hosts?

Ans: MAC addresses for remote hosts are not known on the local network, so the MAC address of the default-gateway is used. After the packet reaches the default-gateway router, the Layer 2 information is stripped from the packet and a new Layer 2 header is attached with the destination MAC address of the next hop router.